* The set $S = \{u_1, u_2, \ldots, u_n\}$ is said to be orthogonal if $\langle u_i, u_j \rangle = 0$ where $i \neq j$.

The set $S = \{u_1, u_2, \ldots, u_n\}$ is said to be orthonormal if (i) $\langle u_i, u_j \rangle = \begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j \end{cases}$

* Consider the set

$\{1, \sin t, \sin 2t, \ldots, \sin nt, \ldots, \cos t, \cos 2t, \ldots, \cos nt, \ldots\}$

in $[-\pi, \pi]$ is orthogonal set.

$$\langle f(t), g(t) \rangle = \int_{-\pi}^{\pi} f(t)\, g(t)\, dt$$

$$\int_{-\pi}^{\pi} \sin nt \cdot \cos mt \, dt, \quad \text{let } m \neq n$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} [\sin(n+m)t + \sin(n-m)t]\, dt$$

$$= \frac{1}{2} \left[\frac{\cos(n+m)t}{n+m} + \frac{\cos(n-m)t}{n-m} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2} \left[\frac{(-1)^{n+m}}{n+m} + \frac{(-1)^{n-m}}{n-m} - \frac{(-1)^{n+m}}{n+m} - \frac{(-1)^{n-m}}{n-m} \right]$$

$$= 0$$

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